

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAMA, BELAGAVI- 590019



A MINI-PROJECT REPORT ON
**“MINIMALLY BIASED MULTIPLIERS FOR APPROXIMATE
INTEGER MULTIPLICATION”**

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF ENGINEERING
In
ELECTRONICS AND COMMUNICATION ENGINEERING

For the academic year 2023-2024

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Department of Electronics & Communication Engineering

CERTIFICATE



A MINI-PROJECT REPORT

On

“MINIMALLY BIASED MULTIPLIERS FOR APPROXIMATE INTEGER MULTIPLICATION”

This is to certify that the project entitled “**Minimally Biased Multipliers for Approximate Integer Multiplication**” is a bonafide work carried out by H Bharath Bhat (1JS21EC052), Hrushik Raj S (1JS21EC058), Karthik V Krishnan (1JS21EC064), in partial fulfillment for the award of degree of **Bachelor of Engineering in Electronics & Communication Engineering of Visvesvaraya Technological University**, Belagavi during the year 2023-24. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the report deposited in the departmental library. The project report has been approved as it satisfies the academic requirements in respect to project work prescribed for the Bachelor of Engineering degree.

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DECLARATION

We **H Bharath Bhat, Hrushik Raj S & Karthik V Krishnan**, of Bachelor of Engineering in Electronics and Communication, JSS Academy of Technical Education, Bengaluru, hereby declare that the mini-project entitled “**Minimally Biased Multipliers For Approximate Integer Multiplication**” has been carried out independently by us at JSS Academy of Technical Education, Bengaluru, under the guidance of **Dr.Thejaswini P**, Designation, Department of Electronics and Communication JSSATEB, Bengaluru.

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ABSTRACT

Approximate multipliers have emerged as a crucial component in modern error-resilient, compute-intensive applications, where reducing area and power consumption is paramount. In this report, we introduce a novel error-configurable, minimally-biased approximate integer multiplier design (MBM). The MBM design incorporates a distinctive error-reduction mechanism, effectively combined with an approximate logarithm-based integer multiplier, to balance the trade-off between accuracy and resource efficiency. Additionally, we propose an optimization strategy for the MBM, which involves eliminating leading-one detection and barrel shifting logic, thus enhancing its applicability in approximate floating-point (FP) multiplier architectures. Beyond this, we also implement one of the existing state-of-the-art approximate multipliers to provide a comprehensive parametric and comparative analysis against our MBM design. This comparative study highlights the strengths and weaknesses of both approaches, offering valuable insights into their performance under various conditions. All multiplier designs were synthesized using the TSMC 45nm standard-cell library, ensuring a fair and consistent evaluation of their area, power, and error metrics.



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Subject: Mini Project Work

Course Outcomes

Subject Code: 21ECMP67

COs	MINIPROJECTWORK	Blooms Level
CO1	Apply the conceptual interpretation of the mathematical principles, knowledge and skills in the field of Electronics and Communication Engineering to formulate a real time problem.	L3, L4
CO2	Analyze the formulated engineering problem and apply appropriate engineering techniques with suitable design process.	L4
CO3	Develop creative solutions with appropriate state-of-the-art techniques/tools.	L4
CO4	Test and validate the solution of the developed prototype against the design specifications.	L4
CO5	Demonstrate the knowledge of professional ethics in documentation and presentation of the work.	L4

This Project maps to the following Program

Outcomes: CO PO Mapping:

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	-	-	2	2	2	2	2	3	2
CO2	-	3	3	-	-	-	-	-	-	-	-	-
CO3	-	-	3	3	3	-	-	-	-	-	-	-
CO4	-	-	-	3	3	-	-	2	-	-	-	-
CO5	-	-	-	-	-	2	2	-	2	2	3	2
CO	3	3	3	3	3	2	2	2	2	2	3	2

CO PSO Mapping:

CO/ PSO	PSO1	PSO2	PSO3
CO1	3	-	3
CO2	3	3	3
CO3	3	3	3
CO4	3	3	3
CO5	3	-	-
CO	3	3	3

CO PO Justification

Cos	POs	Level	Justification
CO1	PO1	3	The Mini project problem formulation needs application of mathematics, science & engineering fundamentals.
	PO6	2	The Mini project work address societal needs, health, safety & legal issues with responsibilities relevant to professional engineering and practice.
	PO7	2	The Mini project work demonstrates the engineering knowledge needed for sustainable development.
	PO8	2	The Mini project work applies ethical principles, professional responsibility in the process of implementation, report preparation & presentation.
	PO9	2	The Mini project work is carried out by group of two or four students which involves individual and teamwork.
	PO10	2	The Mini project work is to be presented orally and exhibited in the exhibition through which the work is communicated effectively. The Mini project work is Documented in the report form based on the instructions and corrections mentioned by the faculty.
	PO11	3	The Mini project work demonstrates project management to execute effectively by the students (Individual & team work). The Mini project work also addresses the financial aspects in its implementation.
	PO12	2	The Mini project work prepares the students for independent and lifelong learning in the context of technological change.
CO2	PO2	3	The Mini project work identifies the issues and formulates the problem through review of research literature. Analyze the engineering problem using principles of mathematics, science& engineering
	PO3	3	The Mini project work has to design or develop solutions for the problems formulated with consideration for health and safety, cultural, societal and environmental needs.
CO3	PO3	3	The Mini project work has to design or develop solutions for the problems formulated with consideration for health and safety, cultural, societal and environmental needs.
	PO4	3	The Mini project work provides valid conclusion using research based knowledge design of experiments analysis and interpretation of data.
	PO5	3	The Mini project work needs application of appropriate techniques resources and modern engineering, IT tools for prediction and modeling of the designed or developed system.
CO4	PO4	3	The Mini project work provides valid conclusion using research based Knowledge design of experiments analysis and interpretation of data.
	PO5	3	The Mini project work needs application of appropriate techniques resources and modern engineering, IT tools for prediction and modeling of the designed or developed system.
	PO8	2	The Mini project work applies ethical principles, professional responsibility in the process of implementation, report preparation & presentation.
CO5	PO6	2	The Mini project work in its documentation & presentation address societal needs, health, safety & legal issues with responsibilities relevant to professional engineering and practice.
	PO7	2	The Mini project work has to document & present the engineering knowledge needed for sustainable development.
	PO8 & 9	2	The Mini project work is carried out by group of two or four students which involves individual and teamwork in preparation of documentation & presentation.

	PO10	2	The Mini project work is to be presented orally and exhibited in the exhibition, through which the work is communicated effectively. The Mini project work is documented in the form of report based on the instructions and corrections mentioned by the faculty.
	PO11	3	The Mini project work demonstrates project management to execute effectively by the students (Individual & teamwork). The Mini project work also addresses the financial aspects in its implementation which are documented in the report and presented.
	PO12	2	The process of Mini project work documentation & presentation prepares the students for independent and lifelong learning in the context of technological change.

CO PSO Justification

Cos	PSOs	Level	Justification
CO1	PSO1	3	The Mini project work aims to solve engineering problem of electronics communication in VLSI Design, Embedded Systems, Communication Engineering, Power Electronics and Control Systems.
	PSO3	3	The Mini project work applies mathematical principles, knowledge and skills of Electronics & Communication Engineering in the designing, realization and validation of various electronic systems.
CO2	PSO1	3	The Mini project work solves engineering problems by using appropriate engineering technique with suitable design process in the field of Electronics & Communication Engineering.
	PSO2	3	To analyze the formulated engineering problem, suitable programming skills and high level languages are necessary.
	PSO3	3	The design of Mini project work needs analysis of the formulated engineering problems.
CO3	PSO1	3	To solve engineering problem in the field of electronics and communication, state-of-the-art, modern tools or technique has to be used in the Mini project work.
	PSO2	3	Programming skills and assembly high level language will help in solving engineering problems.
	PSO3	3	To design, realize and valid at electronics system the programming skills using assembly and high level languages are essential.
CO4	PSO1	3	The designed electronics system has to be tested and validated before any real time implementation of the product.
	PSO2	3	Testing and validation is carried out with programming skills in assembly and high level languages.
	PSO3	3	Electronics system is designed and validated before the product development.
CO5	PSO1	3	The Mini project work solves a problem in Electronics & Communication Engineering needs to be documented and presented for sustainable development of the Mini project work.



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ABSTRACT

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CHAPTER 1

INTRODUCTION

Approximate computing has become a key strategy in the design of energy-efficient and high-performance systems, particularly in error-resilient applications such as multimedia processing and machine learning. These applications often involve computationally intensive operations, with multiplication being a central task that consumes significant resources. By introducing controlled errors in multiplication, it is possible to achieve simpler, faster, and more power-efficient hardware, albeit with some acceptable level of inaccuracy.

Traditional multipliers are designed to perform exact computations, resulting in complex and resource-demanding hardware implementations. In contrast, approximate multipliers modify the standard multiplication algorithm to reduce complexity. This can involve techniques such as omitting less significant bits, simplifying arithmetic operations, or reducing the logic required for certain computational steps. These modifications enable the creation of hardware that is more area-efficient and consumes less power, while still providing outputs that are sufficiently accurate for specific applications.

The core challenge in designing approximate multipliers lies in finding the optimal balance between accuracy and resource efficiency. Different applications have varying tolerances for computational errors, necessitating a diverse design space of approximate multipliers that can offer a range of trade-offs between these factors. The design process must ensure that the error introduced remains within acceptable limits, while still achieving the desired reductions in hardware complexity and power consumption.

One advanced design strategy for approximate multipliers involves the development of minimally biased multipliers (MBMs). These multipliers incorporate error-reduction mechanisms to minimize the bias introduced by approximation, making them suitable for applications where a low level of error is critical. In the context of approximate floating-point multiplication, further optimization can be achieved by removing components such as leading-one detection and barrel shifting logic, which are typically used in exact multipliers but contribute significantly to their complexity. This approach reduces both area and power consumption, while the introduced bias remains minimal and manageable.

1.1 Motivation

The motivation behind implementing the MBM multiplier and conducting a comparative analysis stem from the pressing need to enhance computational efficiency in modern error-resilient systems. As applications in fields like multimedia processing and machine learning become increasingly complex, they demand high-performance computing solutions that can operate within constrained power and area budgets. Approximate multipliers, such as the Minimally Biased Multiplier (MBM), offer a compelling solution by allowing for controlled errors to achieve significant reductions in hardware complexity and power consumption. Implementing the MBM multiplier provides an opportunity to explore its effectiveness in balancing accuracy and efficiency, leveraging its unique error-reduction mechanisms to minimize bias while optimizing performance. The comparative analysis further aims to evaluate the MBM's advantages over existing multiplier designs, assessing its potential benefits in terms of area, power, and error trade-offs. This comparative approach not only validates the performance of the MBM multiplier but also provides valuable insights into its practical applicability and benefits in real-world scenarios, ultimately contributing to the advancement of more efficient and adaptable computing systems.

1.2 Problem Statement

Design and Implement a Minimally Biased Multiplier for Approximate Integer Multiplication.

1.3 Objectives

- Implement Minimally Biased Multipliers (MBM) for Approximate Integer Multiplication
- Perform Error analysis of the MBM multipliers in terms of mean error, error bias and peak error.
- Comparative analysis of MBM with existing approximate multiplier.

CHAPTER 2

LITERATURE SURVEY

The paper titled "**Energy-efficient approximate multiplication for digital signal processing and classification applications**" by S. Narayanamoorthy, H. A. Moghaddam, Z. Liu, T. Park, and N. S. Kim, published in IEEE Transactions on VLSI Systems, explores methods to reduce energy consumption in digital signal processing and classification applications through the use of approximate multipliers. These approximate multipliers trade off some computational accuracy for significant reductions in power usage. The authors present a detailed analysis and demonstrate that these multipliers can be effectively integrated into various digital signal processing and classification tasks, achieving substantial energy savings with minimal impact on overall performance.

The paper "**A survey of techniques for approximate computing**" by S. Mittal, published in ACM Computing Surveys, vol. 48, no. 4, pp. 62:1-62:33, 2016, provides a comprehensive review of approximate computing, which aims to trade off computational accuracy for improved efficiency and reduced resource consumption. The survey discusses various strategies to identify approximable sections of programs and monitor output quality, alongside techniques for applying approximate computing across different processing units such as CPUs, GPUs, and FPGAs. It also examines the implications of approximate computing on processor components and memory technologies, presenting a detailed analysis of how these techniques can meet the growing performance demands within limited resource budgets.

The paper "**Approximate computing and the quest for computing efficiency**" by S. Venkataramani, S. T. Chakradhar, K. Roy, and A. Raghunathan, presented at the 2015 DAC (Design Automation Conference), explores the concept of approximate computing as a means to enhance computing efficiency amidst the diminishing returns from traditional technology scaling. The authors discuss the shift towards multicore and heterogeneous accelerator-based architectures to meet performance demands within constrained power budgets. The paper highlights various principles and techniques of approximate computing, emphasizing their potential to deliver significant improvements in energy efficiency and computational performance by allowing controlled trade-offs in accuracy.

The paper "**Architectural-space exploration of approximate multipliers**" by S. Rehman, W. El-Harouni, M. Shafique, A. Kumar, and J. Henkel, presented at the 2016 ICCAD (International Conference on Computer-Aided Design), introduces a methodology for designing approximate multipliers by exploring a wide architectural space. The methodology generates various design points by adjusting three key parameters: different types of elementary approximate multiplication modules, different types of elementary adder modules for summing partial products, and the selection of bits for approximation in wide-bit multiplier designs. This approach allows for the creation and exploration of a wide range of multiplier designs, balancing trade-offs between computational accuracy and energy efficiency.

The paper "**Reducing power by optimizing the necessary precision/range of floating-point arithmetic**" by J. Y. F. Tong, D. Nagle, and R. A. Rutenbar, published in IEEE Transactions on VLSI Systems, vol. 8, no. 3, pp. 273–286, 2000, explores strategies to reduce power consumption in floating-point (FP) hardware by minimizing the bitwidth of FP data. The study analyzes several FP programs manipulating low-resolution human sensory data, finding that significant reductions in mantissa bitwidth do not impact the accuracy of most programs in the benchmark suite. This approach effectively reduces the power cost of FP operations, making low-power systems more feasible without sacrificing computational accuracy.

The paper "**Power-and area-efficient approximate Wallace tree multiplier for error-resilient systems**" by K. Bhardwaj, P. S. Mane, and J. Henkel, presented at the International Symposium on Quality Electronic Design in March 2014, introduces a new design for a power and area-efficient approximate Wallace Tree Multiplier (AWTM). This design is targeted at error-tolerant applications and aims to optimize Latency-Power-Area (LPA) metrics by using a bit-width aware approximate multiplication algorithm. The paper proposes a carry-in prediction method to reduce the critical path and further enhances efficiency with a hardware-efficient precomputation of carry-in. The resulting design demonstrates significant improvements in both power and area efficiency while maintaining acceptable accuracy for error-resilient systems.

The paper by K. Y. Kyaw, W. L. Goh, and K. S. Yeo, titled "**Low-power high-speed multiplier for error-tolerant application,**" presented at the EDSSC in December 2010, focuses on the design of a multiplier optimized for error-tolerant applications. The authors propose a novel multiplier architecture that achieves both low power consumption and high-speed operation by leveraging error-tolerant techniques. The approach involves trade-offs between accuracy and performance, making it suitable for applications where slight inaccuracies can be tolerated in exchange for significant improvements in energy efficiency and processing speed. The paper demonstrates the effectiveness of the proposed design through experimental results, showing its potential for use in systems where power and speed are critical factors.

In "**Unified Mitchell-based approximation for efficient logarithmic conversion circuit,**" J. Y. L. Low and C. C. Jong tackle the challenge of optimizing logarithmic conversion circuits used in digital systems. The paper introduces a unified approximation technique derived from the Mitchell-based approximation, aimed at enhancing both the efficiency and performance of these circuits. The authors focus on balancing trade-offs between computational accuracy and hardware resource requirements.

Their approach involves a systematic method for approximating logarithmic functions, which is particularly beneficial for applications where exact precision is less critical compared to speed and power efficiency. The paper details how the proposed technique simplifies the logarithmic conversion process while still achieving high accuracy through careful calibration of approximation parameters. The results from their experiments show significant improvements in terms of speed, power consumption, and circuit area compared to traditional methods.

Additionally, the authors provide a comprehensive comparison of their method with existing techniques, illustrating the advantages in practical scenarios. Their unified approach not only enhances the efficiency of logarithmic conversions but also offers a flexible framework that can be adapted to various application requirements, making it a valuable contribution to the field of digital circuit design.

The paper, "**Computation of the base two logarithms of binary numbers**," M. Combet, H. V. Zonneveld, and L. Verbeek address the computation of base-two logarithms for binary numbers, a fundamental operation in digital systems and computer arithmetic. The paper focuses on developing efficient algorithms and methods for calculating logarithms, specifically tailored to binary numbers, which are crucial for various applications in electronic computing.

The authors introduce several innovative approaches to improve the computational efficiency of these logarithmic calculations. They propose algorithms that exploit the binary representation of numbers to simplify and accelerate the logarithmic computation process. The paper provides a thorough examination of these methods, including their theoretical underpinnings, algorithmic structure, and practical implementation details.

Combet, Zonneveld, and Verbeek offer comparative analysis and experimental results demonstrating the performance benefits of their techniques over previous methods. Their work highlights improvements in computational speed and accuracy, which are significant for the design and performance of digital systems. The paper also discusses the implications of these methods for hardware design and electronic computation, emphasizing their practical relevance in enhancing the efficiency of logarithmic operations in digital systems.

Overall, this paper represents a valuable contribution to the field of digital arithmetic, providing a foundation for future research and development in efficient logarithmic computation and influencing subsequent advancements in electronic computing methodologies.

E. L. Hall, D. D. Lynch, and S. J. Dwyer explore ways to enhance the "**Efficiency of digital filtering operations by using approximate binary logarithms**" to simplify multiplication and division tasks. Their approach reduces the complexity of these calculations, which speeds up the processing time in digital filters used for signal processing.

Rather than relying on precise arithmetic, the authors use approximations of binary logarithms to achieve faster computations while maintaining sufficient accuracy for practical applications. They provide detailed explanations of their method and demonstrate its effectiveness through experimental results.

This technique allows for quicker and more efficient digital filtering, which is crucial for real-time signal processing applications. The paper emphasizes the balance between achieving

accuracy and improving computational speed, offering valuable insights for optimizing digital signal processing.

In "**CMOS VLSI Implementation of a Low-Power Logarithmic Converter**," H. Abed and R. E. Siferd focus on developing a logarithmic converter designed to operate with minimal power consumption using CMOS VLSI technology. Their work emphasizes optimizing the converter's performance while reducing energy usage, which is crucial for power-sensitive applications. The paper provides a detailed account of the circuit design, including the specific techniques used to achieve low power operation, such as efficient transistor sizing and careful circuit layout. The authors also include experimental results that validate the effectiveness of their design, showcasing improvements in power efficiency without compromising accuracy or speed. This work represents a significant contribution to the field of low-power electronics, offering a practical approach for implementing logarithmic converters that are both energy-efficient and effective for a range of digital applications.

In the paper, "**VLSI Implementation of a Low-Power Antilogarithmic Converter**," K. H. Abed and R. E. Siferd present an innovative approach to designing an antilogarithmic converter with a focus on minimizing power consumption using VLSI (Very-Large-Scale Integration) technology. The authors detail the architecture and implementation strategies employed to achieve a low-power operation, addressing key challenges in reducing energy usage while maintaining high performance. Their design incorporates several optimizations, including novel circuit techniques and efficient use of CMOS technology, to ensure that the antilogarithmic converter operates with minimal power dissipation. The paper provides comprehensive experimental results demonstrating the effectiveness of their approach, highlighting significant improvements in power efficiency compared to existing solutions. This contribution is particularly relevant for applications where power constraints are critical, and the design methodologies discussed offer valuable insights into achieving energy-efficient VLSI circuits for antilogarithmic functions.

The paper "**Near-Threshold Computing: Reclaiming Moore's Law through Energy Efficient Integrated Circuits**" explores the concept of near-threshold computing (NTC) as a

strategy to improve energy efficiency in integrated circuits, addressing the challenges of power consumption that have arisen as Moore's Law slows down. The authors propose operating circuits at a supply voltage close to the threshold voltage of transistors, significantly reducing power usage while maintaining acceptable performance levels.

While NTC offers substantial energy savings, it comes with challenges such as increased delay variability and reduced clock speed. The paper discusses architectural and circuit-level techniques to address these issues, making NTC viable for energy-sensitive applications like battery-powered devices and data centers. Overall, the work highlights the potential of NTC to extend the benefits of Moore's Law by enabling more energy-efficient integrated circuit designs in the future.

The paper titled "**A 530mV 10-Lane SIMD Processor with Variation Resiliency in 45nm SOI**" presents a detailed design of a low-voltage SIMD (Single Instruction, Multiple Data) processor optimized for energy efficiency and variation resiliency. The authors focus on operating the processor at an ultra-low voltage of 530mV, which is significantly lower than typical operating voltages, to achieve substantial power savings.

The processor is built using 45nm Silicon-On-Insulator (SOI) technology, which is known for its advantages in reducing leakage power and improving performance under lower voltage conditions. The design includes a 10-lane SIMD architecture, which is well-suited for parallel processing tasks, making it efficient for workloads that benefit from high data throughput.

A key innovation discussed in the paper is the implementation of variation resiliency techniques, which address the increased sensitivity to process variations that occur at low voltages. These techniques ensure reliable operation of the processor despite the challenges posed by operating in a low-voltage regime. The paper presents experimental results demonstrating the processor's ability to maintain performance and energy efficiency, making it a promising solution for applications where power consumption is critical, such as in portable and embedded systems. This work contributes to the field by showcasing the potential of low-voltage operation in achieving energy-efficient processing while overcoming the associated technical challenges.

The paper "**Near-Threshold Computing for Very Wide Frequency Scaling: Approximate Adders to Rescue Performance**" explores the use of near-threshold computing (NTC) as a means to achieve energy efficiency in integrated circuits, particularly focusing on the challenges of maintaining performance across a wide range of operating frequencies. The authors propose the use of approximate adders as a solution to the performance degradation typically associated with NTC.

Operating circuits near the threshold voltage of transistors significantly reduces power consumption but can also lead to reduced processing speed and increased susceptibility to variations. To address these issues, the paper introduces approximate adders, which intentionally introduce small errors in calculations to enhance speed and energy efficiency while operating under near-threshold conditions. These adders are designed to strike a balance between accuracy and performance, making them suitable for applications where perfect precision is not essential.

The paper includes experimental results that demonstrate how these approximate adders can effectively mitigate the performance loss typically seen in near-threshold computing, enabling wider frequency scaling while still maintaining low power consumption. This work contributes to the ongoing research in energy-efficient computing by offering a practical approach to preserving performance in NTC environments, particularly in applications where energy efficiency and flexible frequency scaling are critical.

The paper "**ImageNet Classification with Deep Convolutional Neural Networks**" by A. Krizhevsky, I. Sutskever, and G. E. Hinton presents a groundbreaking approach to image classification using deep convolutional neural networks (CNNs). The authors describe the architecture and training of a deep CNN that was used to achieve state-of-the-art results on the ImageNet Large Scale Visual Recognition Challenge (ILSVRC). This dataset, which contains millions of labeled images across thousands of categories, is a benchmark for evaluating the performance of image classification algorithms.

The CNN designed by the authors consists of multiple layers, including convolutional, pooling, and fully connected layers, which are trained end-to-end to optimize classification accuracy. A key innovation in the paper is the use of rectified linear units (ReLUs) as activation functions,

which significantly accelerate the training process compared to traditional activation functions. Additionally, the network employs techniques such as dropout to reduce overfitting and data augmentation to increase the diversity of training data.

The results of their experiments demonstrated a significant improvement in classification accuracy over previous methods, establishing deep CNNs as a powerful tool for image recognition tasks. The success of this model paved the way for the widespread adoption of deep learning techniques in computer vision and beyond. This paper is considered a seminal work in the field of deep learning and has had a profound impact on subsequent research and applications in neural networks and image processing.

The paper "**ImageNet Large Scale Visual Recognition Challenge**" provides a comprehensive overview of the ImageNet Large Scale Visual Recognition Challenge (ILSVRC), a pivotal benchmark in the field of computer vision. The authors, including O. Russakovsky and colleagues, detail the organization, evaluation methodology, and impact of this competition, which has played a crucial role in advancing the state-of-the-art in visual recognition.

The paper outlines the challenge's structure, which involves classifying images into thousands of categories and detecting objects within images, using a dataset that includes millions of labeled images. It describes the evaluation metrics used to measure performance, such as top-1 and top-5 accuracy, and discusses the various approaches and algorithms submitted by participants over the years. Key contributions highlighted include the shift towards deep learning techniques, particularly convolutional neural networks (CNNs), which have significantly improved classification accuracy and driven innovation in the field. The paper also discusses the challenges and limitations faced during the competition, including issues related to dataset biases and the need for more robust evaluation methods.

Overall, this paper serves as a critical resource for understanding the evolution of visual recognition technologies and the impact of the ILSVRC in shaping research and development in computer vision. It provides insights into the progress made in the field and sets the stage for future advancements and challenges.

CHAPTER 3

METHODOLOGY

3.1 Introduction to Multipliers

A multiplier is a fundamental digital circuit used to perform multiplication operations on binary numbers. It is a critical component in various applications, including digital signal processing, arithmetic units in processors, and embedded systems. The primary function of a multiplier is to execute the multiplication of two binary inputs, producing a product that is typically represented as a larger binary number. The design of multipliers in VLSI involves optimizing both the speed and area to ensure efficient performance in integrated circuits. Common types of multipliers used in VLSI include array multipliers, booth's algorithm multipliers, and Wallace tree multipliers, each with its specific trade-offs in terms of complexity, speed, and resource usage.

The implementation of multipliers in VLSI circuits requires careful consideration of several factors to achieve high performance and low power consumption. Array multipliers, for example, are known for their straightforward design and regular structure but can be less efficient in terms of speed and area compared to more advanced techniques like Booth's algorithm or Wallace tree multipliers. Booth's algorithm is particularly effective for multiplying signed numbers and reducing the number of partial products, while Wallace tree multipliers use a tree structure to accelerate the addition of partial products, thus improving speed. VLSI design techniques such as pipelining, parallelism, and optimized logic gate arrangements are often employed to enhance the performance of multipliers. Effective multiplier design is essential for achieving high-speed computation and efficient use of silicon area in modern integrated circuits.

3.2 Minimally Biased Multipliers for Approximate Integer

Minimally biased multipliers for approximate integer and floating-point multiplication are specialized circuit designs aimed at achieving efficient multiplication with a controlled trade-off between accuracy and performance. These multipliers are designed to provide approximate results while minimizing the bias introduced by approximation errors. The key idea is to

balance the reduction in computational complexity and power consumption with the need to maintain acceptable levels of accuracy for practical applications.

3.2.1 Design

The proposed MBM multiplier design aims to approximate the multiplication of two N -bit integers, B_1 and B_2 , using a hardware-efficient approach based on logarithmic approximation. This design reduces complexity and enhances performance by simplifying traditional multiplication operations.

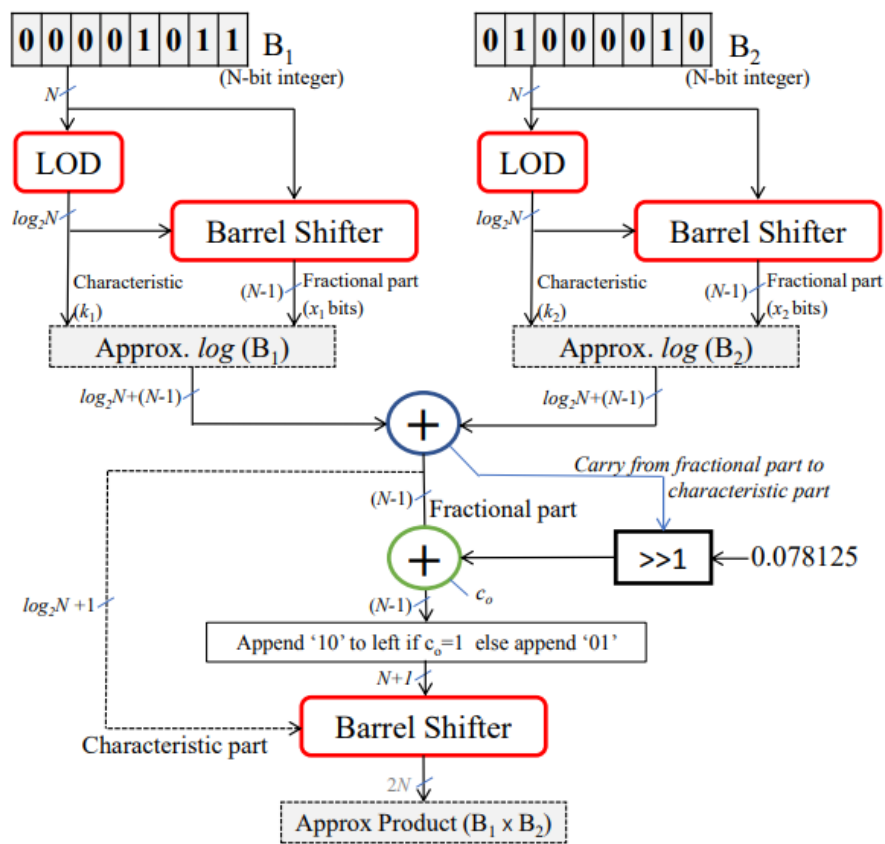


Figure 1, The proposed MBM multiplier design.

Leading-One Detector (LOD):

The Leading-One Detector (LOD) is crucial for identifying the position of the most significant '1' in the binary representation of the input integers, B_1 and B_2 . By detecting the leading '1', the LOD helps in determining the characteristic part of the input, which is a key step in the logarithmic approximation.

The output of the LOD represents the position of the leading '1', which is then used to isolate the characteristic and fractional parts of the input. This step simplifies the subsequent logarithmic transformation by providing a binary logarithmic estimate of the input values.

The characteristic parts, derived from the leading one positions, are essential for the approximate logarithmic representation, facilitating an efficient and simplified multiplication process in the logarithmic domain.

Barrel Shifter:

The Barrel Shifter, controlled by the output of the LOD, shifts the input bits to align the leading '1' at a fixed position. This operation isolates the fractional part of the input, which contains the remaining significant bits after the characteristic part.

By aligning the leading '1' to a predetermined position, the Barrel Shifter ensures that the fractional part is consistently represented, simplifying the hardware required for subsequent processing stages.

The use of a Barrel Shifter helps maintain the precision of the fractional part, which is critical for the accuracy of the logarithmic approximation and the overall multiplication result.

Approximate Logarithm Calculation:

The characteristic parts obtained from the LOD and Barrel Shifter represent the approximate logarithm base-2 values of the inputs, B1 and B2. This approximation leverages the positions of the leading '1' to provide a simplified logarithmic representation.

The fractional parts, isolated by the Barrel Shifter, are combined with the characteristic parts to form a complete logarithmic approximation of the inputs. This combination involves straightforward bit-wise operations, reducing the computational complexity.

The logarithmic approximation effectively transforms the multiplication operation into an addition, significantly simplifying the hardware design by converting the problem into the logarithmic domain.

Addition of Logarithmic Values:

The logarithmic values of B1 and B2, represented by the characteristic and fractional parts, are added together to produce an intermediate sum. This sum represents the logarithm of the product of the original inputs in the logarithmic domain.

The addition process involves handling the bit-wise sum of the logarithmic values, ensuring that the carry bits are appropriately managed to maintain accuracy. The intermediate sum captures the combined logarithmic magnitude of the inputs.

The resulting sum provides a basis for the subsequent anti-logarithmic conversion, where the simplified addition operation replaces the more complex traditional multiplication, reducing hardware complexity and enhancing efficiency.

Error Reduction Term:

To address the approximation error inherent in the logarithmic representation, a small constant value (0.078125) is added to the intermediate sum. This error reduction term compensates for the deviation caused by the logarithmic approximation.

The inclusion of the error reduction term involves careful bit-wise addition, ensuring that the adjustment is accurately integrated into the intermediate sum. This step is critical for improving the precision of the final multiplication result.

The adjustment process includes handling carry bits, which may arise from the addition of the error reduction term. The carry bit is managed separately to ensure that the overall calculation remains accurate and consistent.

Anti-logarithmic Conversion:

The adjusted sum, obtained after adding the error reduction term, is converted back from the logarithmic domain to the normal domain using an anti-logarithmic operation. This conversion is essential for obtaining the final product of the original inputs.

Instead of utilizing resource-intensive components like large barrel shifters, the design employs simpler 2x1 multiplexers for the anti-logarithmic conversion. This choice reduces hardware complexity and enhances the efficiency of the conversion process.

The anti-logarithmic conversion involves minimal shifting operations, primarily determined by the presence of carry bits from the fractional part adjustment. This step ensures that the final product maintains high accuracy while using simplified hardware components.

Output:

The final product, representing the approximate multiplication of B1 and B2, is obtained by combining the adjusted characteristic and fractional parts after the anti-logarithmic conversion.

This output is a $2N$ -bit representation, capturing the result of the logarithmic approximation-based multiplication. The design ensures that the final output maintains a high level of accuracy despite the approximation, leveraging the error reduction term and careful bit-wise operations throughout the process.

By utilizing a logarithmic approximation approach, the MBM multiplier design achieves a balance between hardware efficiency and computational precision, providing a practical and effective solution for approximate multiplication of large integers.

3.3 A Dynamic Range Unbiased Multiplier for Approximate Applications

The novel approach DRUM focuses on creating a multiplier which has unbiased error distribution, where the errors cancel out on repeated computations instead of accumulating.

3.3.1 Design

Before delving into the DRUM design, cover some basics of the building blocks of DRUM which will be used in later sections to develop the DRUM design.

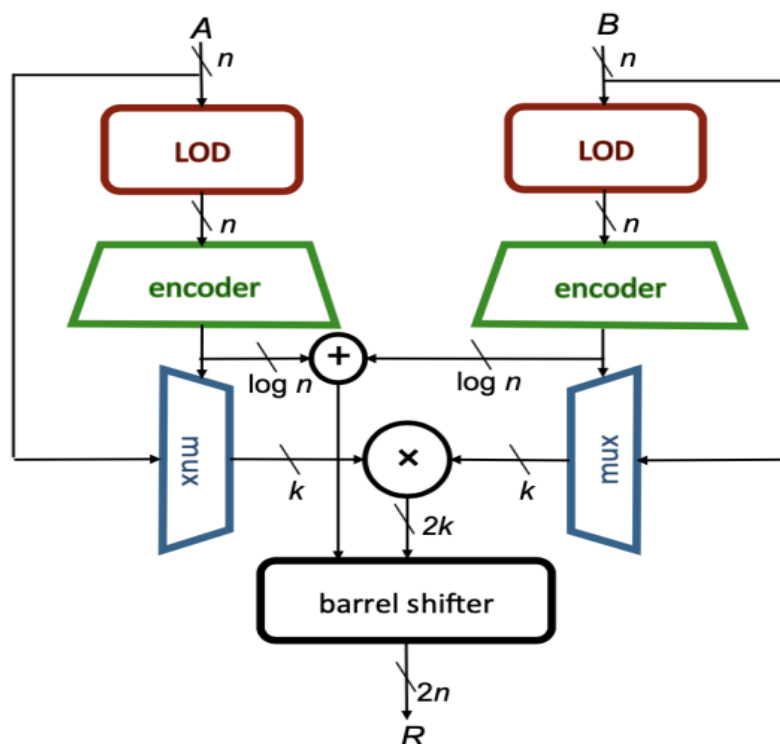


Figure 2 The simplified schematics for DRUM

Leading One Detector (LOD)

The Leading One Detector (LOD) is a critical component in the DRUM design that identifies the position of the most significant '1' bit in a binary number. This detection is essential for focusing the multiplication process on the most significant bits, thereby reducing the computational load and enhancing efficiency.

In the context of an 8-bit input, the LOD scans the binary number from the most significant bit (MSB) to the least significant bit (LSB) to locate the first occurrence of '1'. This position information helps in dynamically adjusting the range of significant bits that will be involved in the multiplication process.

The accuracy and speed of the LOD directly impact the overall performance of the DRUM. An efficient LOD design ensures minimal delay, enabling the DRUM to achieve high-speed operations suitable for real-time applications.

Priority Encoder

The priority encoder takes the output from the LOD, which is a binary number with a single '1' at the position of the most significant bit, and converts it into a binary code. This binary code represents the exact position of the leading '1', which is crucial for further processing in the multiplier.

For an 8-bit input, the priority encoder produces a 3-bit binary code corresponding to the position of the detected '1'. This encoding step compresses the position information into a compact form that can be easily utilized by subsequent blocks such as the multiplexer and the barrel shifter.

The priority encoder must be designed to handle inputs quickly and accurately, ensuring that the encoded position is available with minimal delay. This fast encoding process is essential for maintaining the overall speed and efficiency of the DRUM.

Multiplexer (MUX)

The multiplexer (MUX) in the DRUM design selects specific bits from the input operands based on the position information provided by the priority encoder. By focusing on the bits around the most significant '1', the MUX effectively reduces the number of bits involved in the multiplication, thereby decreasing the computational complexity.

In an 8-bit DRUM, if the core multiplier width k is smaller than 8, the MUX dynamically selects $k - 2$ bits starting from the most significant '1' identified by the LOD. This selective bit extraction ensures that the most relevant bits are used in the multiplication, which helps in maintaining a balance between accuracy and efficiency.

The MUX must be capable of rapidly switching between different bit positions as directed by the priority encoder. This dynamic selection capability is vital for the adaptive nature of the DRUM, allowing it to handle varying input patterns effectively.

Wallace-Tree Multiplier

The Wallace-Tree Multiplier is a high-speed parallel multiplier that forms the core of the DRUM's multiplication process. It takes the selected bits from the MUX and performs the multiplication operation, generating partial products that are then efficiently reduced to produce the final product.

The Wallace-Tree architecture is designed to minimize the number of sequential addition steps required to combine the partial products, leveraging parallel processing to achieve high-speed multiplication. This structure is particularly advantageous in approximate computing, where speed is often prioritized over exact precision.

The design of the Wallace-Tree Multiplier within the DRUM must ensure that it handles the dynamically selected bits with minimal delay, maintaining the overall performance of the multiplier. The use of parallel reduction techniques in the Wallace-Tree Multiplier contributes significantly to the DRUM's ability to perform rapid and efficient multiplications.

Barrel Shifter

The barrel shifter in the DRUM adjusts the output of the Wallace-Tree Multiplier according to the position of the most significant '1' detected by the LOD. This shifting operation is crucial for aligning the multiplication result correctly within the desired output range, ensuring that the final output maintains its significance.

The barrel shifter receives the product from the Wallace-Tree Multiplier and shifts it left or right based on the position information encoded by the priority encoder. This dynamic shifting helps in restoring the positional value of the bits that were initially selected by the MUX, ensuring that the final result is appropriately scaled.

The design of the barrel shifter must accommodate rapid shifting operations with minimal delay. Its ability to quickly adjust the product ensures that the DRUM can deliver accurate results in real-time applications, making it a critical component for maintaining the overall efficiency and accuracy of the multiplier.

Steering Logic

The steering logic in the DRUM coordinates the interactions between the LOD, priority encoder, MUX, Wallace-Tree Multiplier, and barrel shifter. It ensures that each component receives the correct signals and data at the right time, orchestrating the entire multiplication process seamlessly.

This logic block is responsible for managing the flow of information from the detection of the most significant '1' to the final adjustment of the product. It ensures that the LOD's output is correctly encoded, the MUX selects the appropriate bits, and the barrel shifter adjusts the result as needed.

Efficient steering logic design is essential for maintaining the synchronization and timing of the DRUM's operations. It must handle control signals and data paths with precision, ensuring that the entire multiplication process is executed smoothly and rapidly.

CHAPTER 4

RESULTS & DISCUSSIONS

4.1 MBM

The results show the output of an MBM multiplier designed where "Input 1" and "Input 2" are multiplied using both accurate and approximate methods. The "Accurate Product" column represents the exact multiplication results, while the "Approximate Product" column shows the results from the MBM algorithm. The "Error Percent" indicates the relative error between the approximate and accurate products. The results demonstrate that the MBM approach introduces a small error, typically within a range of 5% to 8%, depending on the input values.

Table 1 Simulation Results of MBM

Input 1	Input 2	Accurate Product	Approximate Product	Error Product
32	27	864	904	4.63
44	241	10604	10624	0.189
45	8	360	380	5.556
70	230	16100	16128	0.174
80	249	19920	20224	1.526
92	18	1656	1680	1.449
106	241	25546	25856	1.213
120	21	2520	2512	0.317
158	104	16432	15872	3.408
175	75	13125	13248	0.937
188	73	13724	13824	0729
200	107	21400	20864	2.505
225	29	6525	6592	1.027
254	186	47244	48384	2.413
254	187	47498	48640	2.404
254	188	47752	48896	2.396

This suggests that the MBM multiplier achieves a balance between performance and accuracy, making it a suitable candidate for applications where slight inaccuracies are acceptable in exchange for efficiency.

Leading One Detector:



Figure 3 Simulation Output of LOD

Barrel Shifter:

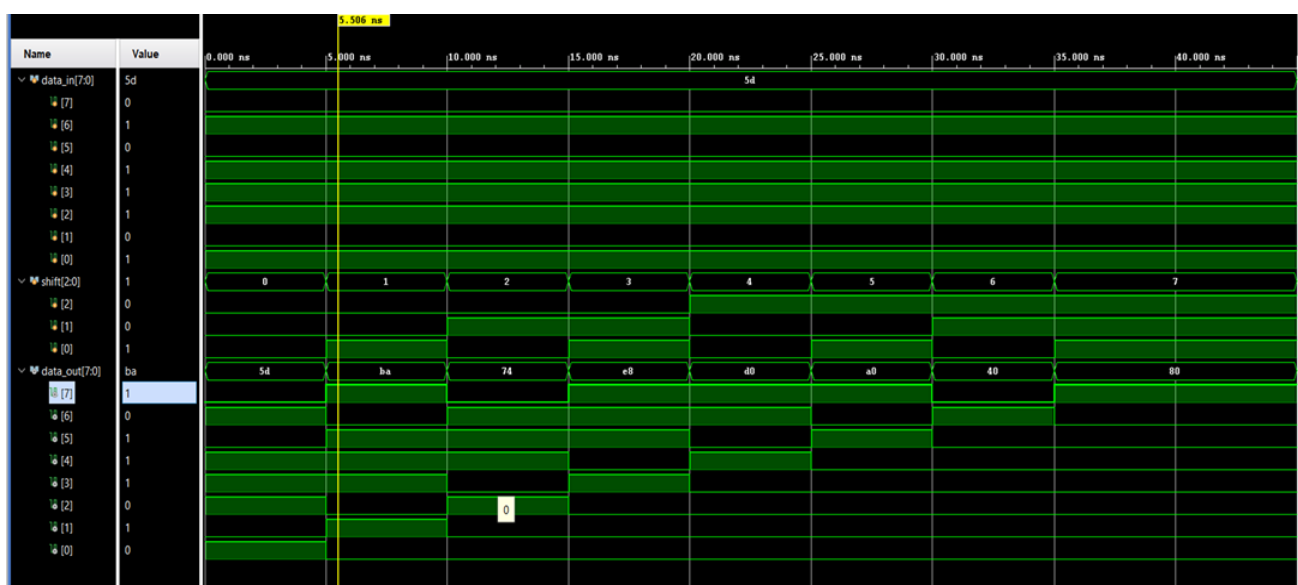


Figure 4 Simulation Output of Barrel Shifter

MBM final Output:



Figure 5 Simulation output of MBM

4.2 DRUM

The results show the output of an MBM multiplier designed using Verilog. Xilinx vivado tool is used for simulation.



Figure 6 Simulation output of DRUM

CHAPTER 5

CONCLUSION AND FUTURE SCOPE

We implemented an error-configurable minimally-biased approximate integer multiplier (MBM) leveraging leading one detectors and barrel Shifters. A novel optimized technique proposed in the paper was used for the approximate integer multipliers (MBM, DRUM, SSM, and MA). Our simulation results show that the proposed MBM integer design provides Pareto optimal design points in the integer multiplier design space of power and area with respect to three different error metrics. MBM can offer better trade-offs than the traditional precision scaling methods while offering much higher area savings ($>25\times$) and power savings ($>50\times$) compared to IEEE single-precision multiplier. The application-level evaluations of the proposed approximate integer and FP multipliers demonstrate minimal degradation in the application-level accuracy.

CHAPTER 6

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